

Xinlei Li

Seattle, WA | 206-251-8801 | xinleijonathan.li@gmail.com | www.linkedin.com/in/1xl2001 | github.com/Xinleilijonathan

EDUCATION

Northeastern University

Seattle, WA

Master of Science, Electrical and Computer Engineering (Computer Science Engineering) | 3.87/4.0

Sep 2024 - Dec 2026

Popular Votes Award & Judges' Choice Award | 2025 Qualcomm & Microsoft On-Device AI Builders Hackathon

MySQL 8.0 Database Developer Oracle Certified Professional

Coursework: Machine Learning & Pattern Recognition, Computer Vision, Robotics Sensing & Navigation

Wenzhou-Kean University

Wenzhou, China

Bachelor of Computer Science, Computer Science

Sep 2020 - Jun 2024

Coursework: Data Structures, Computer Architecture, Operating Systems

SKILLS

Robotics: ROS 2, MoveIt 2, UR RTDE, Isaac Sim / Isaac Lab, Lula, LeRobot, UR12e / UR10 / UR5, Robotiq Hand-E, DYNAMIXEL

Robot Learning & CV: PyTorch, JAX, OpenCV, YOLOv8, NVIDIA Cosmos3-Nano, VLA, imitation learning, action normalization

Sensors & Data: RealSense D435, OAK-D, multi-view RGB, joint states, gripper actions, camera calibration, LeRobot datasets

Programming & Systems: Python, C/C++, Java, Go, SQL, Bash, Linux, Git, Docker, Kubernetes, SLURM / HPC

ROBOTICS RESEARCH

Northeastern University

Seattle, WA

Graduate Researcher - Real-World VLA Adaptation on UR12e

2026 - Present

- Building an **end-to-end robot-learning pipeline** from a low-cost GELLO-style leader arm to dual-RGB, robot-state, gripper, and language demonstrations in **LeRobot** format, followed by VLA fine-tuning and real-robot evaluation on a UR12e with Robotiq Hand-E.
- Built and calibrated a custom **DYNAMIXEL leader arm** with joint-to-joint mapping, motor offset/sign alignment, joint-limit handling, home-pose synchronization, and gripper control for repeatable teleoperation.
- Designed controlled experiments across **20 / 50 / 100 / 200 demonstrations** and pick-place, bolt-twist, and peg-insertion tasks to study sample efficiency, teleoperation quality, and robustness under distribution shift.
- Developing synchronized multi-camera collection and calibration using **RealSense D435 and OAK-D** with UR RTDE state/action logging for reproducible VLA training and evaluation.

SELECTED ROBOTICS PROJECTS

Simulation-Based Vision-Language-Action Policy Development

2026

- Fine-tuned **NVIDIA Cosmos3-Nano** on 250 Berkeley UR5 bottle episodes (21K valid 16-step windows) on H200 GPUs and built a 10D action adapter for translation, 6D rotation, and gripper control.
- Implemented Berkeley-specific action normalization, reducing overall action MAE from **0.3359 to 0.0253** (~92%); translation MAE fell from 0.3266 m to 0.0114 m and rotation6D MAE from 0.3055 to 0.00125.
- Integrated policy actions with **Isaac Sim UR10 + Lula FK/IK**; corrected a 90.04° frame mismatch, achieved 0.352 mm single-action EEf error, and completed a verified bottle-to-pot pick-and-place baseline.

Autonomous Robotic Packing System

2026

- Configured **UR12e motion planning in ROS 2 / MoveIt 2**, brought up real hardware via RTDE, and integrated eye-tracking input for multimodal target selection in a perception-to-manipulation packing pipeline.

WORK EXPERIENCE

Beltways Inc.

Hebron, KY

Software Engineering Intern

Sep 2025 - Dec 2025

- Designed a **high-throughput Python / TypeScript sensor-data pipeline** processing 10M+ signals per second; applied aggregated windowing and standard scaling while preserving key equipment-health indicators.
- Developed an **end-to-end ML hardware-health monitoring system** with anomaly detection and proactive alerting; containerized services with Docker / Kubernetes for scalable deployment.

Shandong Luxin Investment Holding Group Co., Ltd

ShanDong, China

Software Engineering Intern

May 2023 - Aug 2023

- Built ERP / MES services for **manufacturing automation** using Spring Boot and Vue.js; improved backend response time by **40%** through Redis caching and SQL query optimization.
- Designed **RESTful microservices** for quality-control workflows, improving system modularity and contributing to a 7% reduction in manufacturing defect rates.

PUBLICATIONS

An Enhanced YOLOv8 for Flame and Smoke Detection with Dilated Convolution and Image Dehazing

Third Author

- Co-developed **FSD-YOLOv8** by integrating AOD-Net image dehazing and dilated convolutions into YOLOv8 to improve flame/smoke detection in fog- and haze-ambiguous scenes.
- Evaluated on **95,314 FASDD_CV images** and achieved **0.92 mAP / 0.91 F1** versus 0.85 / 0.84 for baseline YOLOv8; validated generalization on Fire-Net and Wildfire Smoke & Fire datasets.